

wavelengths (colors) are spread out in sequence. The term is now also applied to other electromagnetic radiation, and also to refer to characteristic patterns of radiation given off by particular sources.

speed The rate at which something is moving. See also *velocity*.

sperm A male sex cell (*gamete*) that can move to locate a female cell. All animals and some lower plants produce sperm.

spherical trigonometry Trigonometry modified to apply to the surface of a sphere rather than a flat surface.

sporozoite A stage in the life cycle of some microscopic parasites.

square root A number which when multiplied by itself yields a given number.

stade (1) An ancient Greek unit of measurement. (2) A period of geological time when glaciers have stopped retreating.

stamen The male organ of a flower. See also *anther*.

standard model The principle theoretical framework of particle physics, combining theories on how three of the four fundamental forces interact (electromagnetism and the strong and weak nuclear forces) with 12 basic particles (six quarks and six leptons).

star A huge luminous ball of ionized gas (plasma) whose energy emissions are powered by nuclear reactions in its core.

static electricity Phenomena involving nonmoving electric charges on objects.

stem cell A type of cell in the body that is able to divide and grow into other more specialized cells.

sterilization (1) Giving special treatment to equipment to kill life forms such as harmful bacteria. (2) Rendering an animal infertile by performing an operation, using radiation, etc.

stethoscope A diagnostic instrument used to listen to sounds within the body, especially the chest.

stigma The top of the female part of a flower (pistil). It is usually sticky to receive pollen.

stratigraphy The study of rock layers.

stratopause The boundary between the *stratosphere* and *mesosphere*.

stratosphere The part of Earth's atmosphere between the *troposphere* and *mesosphere*.

stratus cloud Usually low cloud and in the form of flat sheets, often bringing light rain.

style The stalk that supports a stigma.

subatomic particle A particle smaller than an atom or its nucleus, such as a proton, neutron, or electron.

subduction boundary A boundary between two tectonic plates in the deep ocean where one plate is being pushed beneath the other.

submersible A usually small vessel for underwater exploration.

substance Any kind of matter.

sunspot A region of the Sun's surface whose temperature is temporarily lowered and which therefore appears darker than its surroundings in images.

superconductivity Phenomenon in which some substances lose all electrical resistance close to *absolute zero*.

supernova An enormous explosion taking place at the end of the life of a very large star.

supersonic Faster than the speed of sound.

surface tension The effect that makes a liquid seem as though it has an elastic "skin," caused by cohesion between the surface molecules.

suspension A mixture of tiny solid particles or globules of liquid in a surrounding medium.

switch A device that turns an electric current on or off, or more generally changes something from one state to another.

synapse A junction between two nerve cells, or between a nerve cell and a muscle or gland cell.

synthesis The combining of separate parts or different theories to make a whole.

taxonomy The classification of living things, and also the principles behind classification.

tectonic Relating to the structure of Earth's crust and its movements. See also *plate tectonics*.

telophase The final stage of *mitosis* and of each part of *meiosis*, in which a nuclear membrane forms around each set of separated *chromosomes*.

temperature A measure of how hot or cold something is.

theodolite A surveying instrument that measures angles using a rotating telescope.

theorem A mathematical rule or statement, especially a truth that is not self-evident but which can be proved by reasoning.

thermal (1) Relating to heat [adjective]. (2) A current of rising hot air in the atmosphere.

thermodynamics The branch of physics dealing with the relationship between heat and other forms of energy.

thermoelectric effect Any of various effects that can occur when there are temperature differences in an electric circuit.

thermosphere The layer of Earth's atmosphere above the *mesosphere*.

three dimensions (3-D) Length, breadth, and depth.

tissue Living material made up of broadly similar kinds of cells and performing a particular function: for example, nerve tissue, muscle tissue.

topography The study of landforms.

torque A twisting force.

trace elements Elements that are needed in only minute amounts by living things.

trade wind A wind that blows from the southeast or northeast all the year round in equatorial regions.

transformer A device that increases voltage while decreasing current, or vice versa. It works only with alternating currents.

transfusion The transfer of blood from a donor to a recipient.

transistor A semiconductor electronic device that acts as a switch, amplifier, or rectifier.

translocation (1) A situation where part of one chromosome has moved to another location, either on the same chromosome or a different one. (2) Movement of materials through a plant.

transmission The conveying of something from one place to another.

transmutation (1) The evolutionary change of one species into another. (2) The conversion of one kind of atom to another through a nuclear reaction.

transparent Allowing light or other radiation to pass through.

transpiration The loss of water from the surfaces of a plant, especially from its leaves.

transplant A tissue or organ taken from one part of the body to be placed in another part, or in another individual.

transverse wave A wave whose to-and-fro movement takes place at right angles to the line that the wave is traveling. Light waves are an example.

triangulation A method of surveying by measuring angles and distances using the mathematical properties of triangles.

trigonometry The branch of mathematics dealing with calculations involving the sides and angles of triangles.

tropical Relating to the warm regions of Earth that lie between the Tropic of Cancer (23.5° north of the equator) and the Tropic of Capricorn (23.5° south of the equator), or to climates typical of those regions.

tropopause In Earth's atmosphere, the boundary between the *troposphere* and *stratosphere*.

troposphere The lowest layer of Earth's atmosphere, starting from ground level, where most weather events take place.

tsunami A water wave, sometimes of huge size, generated by an earthquake, underwater landslide, or other major disturbance.

turbine A rotating wheel driven by water, or moving gas or air, in order to provide power.

ultrasound Sound with a frequency above that which the human ear can detect.

ultraviolet A type of electromagnetic radiation with wavelengths shorter than visible light.

uncertainty principle The quantum physics principle that it is impossible to measure both the position and momentum of objects at the subatomic level exactly, because the observation of one changes the other.

Universe Traditionally, a term for the totality of everything that exists. Now often used for everything created as a result of the Big Bang, which leaves open the possibility of other universes existing.

urine A fluid that most animals discharge to get rid of waste products and excess water.

vaccine A special preparation of substances that stimulate an immune response, used for inoculation.

vacuum A space in which there is no matter. (In real-life examples a vacuum is only approximate.)

vacuum tube A sealed tube, usually glass, from which most air has been removed and which contains electrodes. Application of electricity causes a beam of electrons to be emitted from the negative electrode (cathode). It is a general term and includes various devices used or formerly used in electronics. Also called an electron tube.

valency The number of chemical bonds that an atom can make with another atom.

valve A device or structure that restricts the flow of fluid or electricity to one direction. In electrical contexts it refers to a kind of *vacuum tube*.

Van der Waals bond A relatively weak kind of chemical bond.

vascular circulation The circulation of blood through blood vessels (arteries, capillaries, and veins) and back to the heart.

vein A blood vessel leading back toward the heart. See also *vascular circulation*.

velocity *Speed* in a particular direction.

Vernier scale A small movable, graduated scale added to a larger scale for increased accuracy in precision measuring instruments, named after Pierre Vernier.

virus (1) Tiny parasitic noncellular life forms consisting mainly of genes with a protective coat that are able to hijack living cells to make copies of themselves. (2) A piece of computer software that can spread itself through computer systems in a manner similar to a biological virus.

visible light Electromagnetic radiation with wavelengths that can be detected as light by the eye.

vitamin Any of various organic compounds needed in small amounts in the diet to preserve health.

viviparous Giving birth to live young as distinct from eggs.

voltage See *potential difference*.

voltaic cell A device that converts the energy of a chemical reaction directly into electricity; colloquially, a battery. See also *battery*.

volume (1) The amount of space something takes up. (2) The loudness of a sound.

vulgar fraction A fraction expressed as one number divided by another rather than using decimal points.

wave A regular oscillation in intensity or concentration. Waves typically travel in a particular direction or directions and transmit energy in that direction.

wavelength The distance between each successive peak or crest of a series of waves.

weak force The force in atomic nuclei responsible for beta decay. It is so called in comparison with the strong force.

weak interaction Another name for the weak force.

white dwarf A small, faint, very dense star, thought to represent the final stage of evolution of stars below a certain mass.

World Wide Web A vast network on the *Internet* for gathering and exchanging data and documents through hypertext links.

worm gear A gear arrangement in which one of the gears is a cylinder with grooves cut in it.

X-ray diffraction A technique of beaming X-rays at materials to obtain information about their internal structure by analyzing the diffraction patterns produced; also called X-ray crystallography.

X-rays A type of high-energy, high-frequency electromagnetic radiation.